Day 105 | Unit 07 | Teacher plan

Understand and Use Percent

Unit 7 Test

Standards: 6.RP.A.1, 6.RP.A.3a, 6.RP.A.3c

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2, 6

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Directions

The first 4 questions (20%) review Unit 6. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

A tray has 6 red tiles and 6 blue tiles. Write the red-to-blue ratio in simplest form.

02 / Previous unit review | Constructed response

A mix uses 5 cups of oats for 6 cups of seeds. Complete the table (oats, seeds): (5, 6), (10, ___), (10, ___).

03 / Previous unit review | Constructed response

Mix A has 5 cups juice for 6 cups water. Mix B has 14 cups juice for 12 cups water. Which is stronger in juice? Show equivalent ratios.

04 / Previous unit review | Constructed response

The ratio of oak trees to pine trees is 2:6. There are 32 trees altogether. Draw a tape diagram and find each count.

05 / Constructed response

Write 50% as a simplified fraction and decimal.

06 / Constructed response

Write 10% as a simplified fraction and decimal.

07 / Constructed response

Write 300% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

08 / Constructed response

Find 40% of 100.

09 / Constructed response

Write 5% as a simplified fraction and decimal.

10 / Constructed response

Find 15% of 160.

11 / Constructed response

Find 40% of 60.

12 / Multiple select

100 is 125% of an unknown quantity w. Select all equations that find w.
A. w = 100 × (125/100)
B. w = 100 + 125
C. (125/100)w = 100
D. w = 100 ÷ (125/100)

13 / Multiple select

20 is 20% of an unknown quantity w. Select all equations that find w.
A. w = 20 × (20/100)
B. (20/100)w = 20
C. w = 20 + 20
D. w = 20 ÷ (20/100)

14 / Multiple select

15 is 25% of an unknown quantity w. Select all equations that find w.
A. w = 15 × (25/100)
B. w = 15 + 25
C. w = 15 ÷ (25/100)
D. (25/100)w = 15

15 / Numeric entry and explanation

A quantity changes from a reference whole of 60 to a compared amount of 75. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

16 / Multiple select

12 is 15% of an unknown quantity w. Select all equations that find w.
A. (15/100)w = 12
B. w = 12 + 15
C. w = 12 ÷ (15/100)
D. w = 12 × (15/100)

17 / Multiple select

24 is 40% of an unknown quantity w. Select all equations that find w.
A. (40/100)w = 24
B. w = 24 × (40/100)
C. w = 24 + 40
D. w = 24 ÷ (40/100)

18 / Multiple select

48 is 60% of an unknown quantity w. Select all equations that find w.
A. (60/100)w = 48
B. w = 48 ÷ (60/100)
C. w = 48 × (60/100)
D. w = 48 + 60

19 / Multiple select

Select all representations of 250%.
A. 250
B. 250/100
C. 250/10
D. 2.5

20 / Multiple select

Select all representations of 350%.

A. 350/10

B. 350

C. 3.5

D. 350/100

Day 105 | Unit 07 | Answer key

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Standards: 6.RP.A.1, 6.RP.A.3a, 6.RP.A.3c

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 1:1

Keep red first; divide 6:6 by 6.

6.RP.A.1 | Constructed response

2. 12; 12

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

3. Mix B

For 12 cups water, A uses 10 cups juice; B uses 14. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

4. 8 oaks and 24 pines

8 equal sections total; each section is 32 ÷ 8 = 4.

6.RP.A.3a | Constructed response

5. 1/2; 0.5

50% = 50/100 = 1/2; divide by 100 for 0.5.

6.RP.A.3c | Constructed response

6. 1/10; 0.1

10% = 10/100 = 1/10; divide by 100 for 0.1.

6.RP.A.3c | Constructed response

7. 3; 3

300% = 300/100 = 3. Dividing by 100 gives 3, which exceeds 1 because 300 exceeds 100.

6.RP.A.3c | Constructed response

8. 40

40/100 × 100 = 40.

6.RP.A.3c | Constructed response

9. 1/20; 0.05

5% = 5/100 = 1/20; divide by 100 for 0.05.

6.RP.A.3c | Constructed response

10. 24

15/100 × 160 = 24.

6.RP.A.3c | Constructed response

11. 24

40/100 × 60 = 24.

6.RP.A.3c | Constructed response

12. C, D - (125/100)w = 100; w = 100 ÷ (125/100)

The part is percent rate times whole. Divide 100 by 5/4; w = 80. Check by multiplication.

6.RP.A.3c | Multiple select

13. B, D - (20/100)w = 20; w = 20 ÷ (20/100)

The part is percent rate times whole. Divide 20 by 1/5; w = 100. Check by multiplication.

6.RP.A.3c | Multiple select

14. C, D - w = 15 ÷ (25/100); (25/100)w = 15

The part is percent rate times whole. Divide 15 by 1/4; w = 60. Check by multiplication.

6.RP.A.3c | Multiple select

15. 125%; denominator 60

Percent = (75 ÷ 60) × 100 = 125%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

16. A, C - (15/100)w = 12; w = 12 ÷ (15/100)

The part is percent rate times whole. Divide 12 by 3/20; w = 80. Check by multiplication.

6.RP.A.3c | Multiple select

17. A, D - (40/100)w = 24; w = 24 ÷ (40/100)

The part is percent rate times whole. Divide 24 by 2/5; w = 60. Check by multiplication.

6.RP.A.3c | Multiple select

18. A, B - (60/100)w = 48; w = 48 ÷ (60/100)

The part is percent rate times whole. Divide 48 by 3/5; w = 80. Check by multiplication.

6.RP.A.3c | Multiple select

19. B, D - 250/100; 2.5

Percent means per 100. Divide 250 by 100 to get 2.5; simplify 250/100 to 5/2.

6.RP.A.3c | Multiple select

20. C, D - 3.5; 350/100

Percent means per 100. Divide 350 by 100 to get 3.5; simplify 350/100 to 7/2.

6.RP.A.3c | Multiple select